Semester I **B. Tech CSE (AI)**

BEE 102: BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LAB

0L + 0T + 4P + 2C **MM 100**

Electrical Lab:

1. To verify Ohm's Law
2. Verification of Kirchoff's Law applied to the DC circuits.
   (a) Parallel and Series combination
   (b) Identification of node points
   (c) Algebraic sum of current at node points
   (d) Algebraic sum of e.m.f.s and voltage drops.
3. To construct a RLC series circuit and measure its impedance, inductive (X_L) and capacitive reactance (X_C), measure phase angle between voltage and current.
4. Make house wiring including earthing for 1-phase energy meter, MCB, ceiling fan, tube light, three pin socket and a lamp operated from two different positions.
5. Study the construction & basic working of ceiling fan. Connect ceiling fan along with regulator through auto-transformer to run and vary speed.
6. Study the construction and connection of single-phase transformer and auto-transformer. Measure input and output voltage and fin turn ratio.
7. Study the construction, circuit, working and application of the following lamps.
(a) Fluorescent Lamp, (b) Sodium Vapour Lamp (c) Mercury vapour lamp (d) Halogen Lamp

Electronics Lab:

1. Study the following devices:
(a) Analog & digital multimeters (b) Function/ Signal generators
(c) Regulated d. c. power supplies (constant voltage and constant current operations)
(d) Study of analog CRO, measurement of time period, amplitude, frequency & phase angle using Lissajous figures.
2. Identification, testing and applications of Resistors, Inductors, Capacitors, PN-diode, SCR,TRAIC, Photo Diode, Zener diode, LED, LCD, BJT, Photo Transistor.
3. Plot V-I characteristic of P-N junction diode & Zener diode & calculate cut-in voltage, reverse Saturation current and static & dynamic resistances. Application of Diode as clipper & clamper.
4. Plot input and output characteristics of BJT in CE configurations. Find its h parameters. Plot gain- frequency characteristic of emitter follower & find out its input and output resistances.
5. Study half wave rectifier and bridge rectifier and effect of filters on wave. Also calculate theoretical & practical ripple factor.